Solution

angle $\theta = 1^\circ\ 55' = 115'$

$= (115 \times 60)'' \times (4.85 \times 10^{-6})$ rad

$= 3.34 \times 10^{-2}$ rad

since $1'' = 4.85 \times 10^{-6}$ rad

Radius of the Earth $= 6.4 \times 10^6 m$

From the Figure 1.5, AB is the diameter of the Earth (b)= $2 \times 6.4 \times 10^6 m$ Distance of the Moon from the Earth x = ?

$$x = \frac{b}{\theta} = \frac{2 \times 6.4 \times 10^6}{3.34 \times 10^{-2}}$$

$$x = 3.83 \times 10^8 \text{m}$$

RADAR method

The word RADAR stands for radio detection and ranging. A radar can be used to measure accurately the distance of a nearby planet such as Mars. In this method, radio waves are sent from transmitters which, after reflection from the planet, are detected by the receiver. By measuring, the time interval (t) between the instants the radio waves are sent and received, the distance of the planet can be determined as

Speed = distance travelled / time taken
(Speed is explained in unit 2)
Distance(d) = Speed of radio waves × time taken

$$d = \frac{v \times t}{2}$$

where v is the speed of the radio wave. As the time taken (t) is for the distance covered during the forward and backward path of the radio waves, it is divided by 2 to get the actual distance of the object. This method can also be used to determine the height, at which an aeroplane flies from the ground.

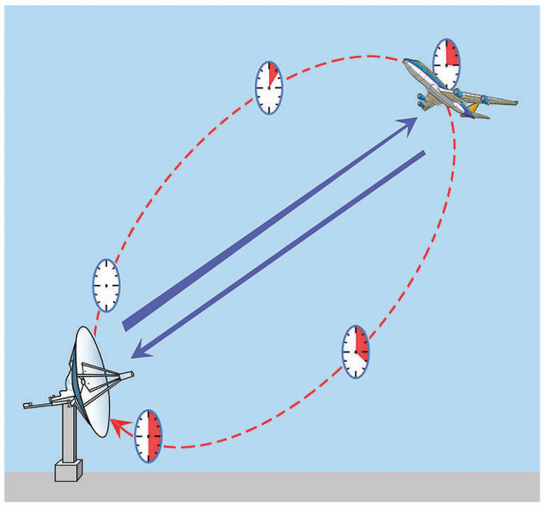


Figure 1.6 RADAR method

EXAMPLE 1.3

A RADAR signal is beamed towards a planet and its echo is received 7 minutes later. If the distance between the planet and the Earth is 6.3×10^{10} m. Calculate the speed of the signal?

Solution

The distance of the planet from the Earth d = 6.3×10^{10} m

Time t = 7 minutes = 7×60 s.
the speed of signal v = ?

The speed of signal

$$v = \frac{2d}{t} = \frac{2 \times 6.3 \times 10^{10}}{7 \times 60} = 3 \times 10^8\ ms^{-1}$$

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